IE410: Introduction to Robotics

Project Report – Part A

Student Details

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Introduction

This report presents the implementation and results of the robotics project for the course **IE410: Introduction to Robotics**. The project focuses on object manipulation using robotic arms equipped with two-finger grippers. The project includes pre-programmed pick-and-place operations, camera-assisted object detection, and coordinated object handover between two robotic arms.

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1. Project Objectives

- Understand the kinematic control of robotic manipulators.
- Implement pick-and-place tasks using pre-planned robot motions.
- Integrate visual sensing using a webcam or smartphone camera.
- Develop object handover coordination between two robotic arms.

- Gain practical experience in robotic motion planning and manipulation.

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2. Task 1 – Pre-Programmed Pick and Place

In this task, the Braccio robotic arm performs a simple pick-and-place operation. The robot grasps an object from Point A and transfers it to Point B using predefined coordinates. The complete motion sequence is programmed offline and executed repeatedly without visual feedback.

Working Procedure

1. Place the object at Point A.
2. Move the robotic arm toward the object.
3. Close the gripper to grasp the object.
4. Move the arm toward Point B.
5. Release the object at Point B.
6. Return the arm to the initial position.

Result

The robotic arm successfully transferred the object from Point A to Point B repeatedly with stable performance.

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3. Task 2 – Camera-Assisted Pick and Place

This task introduces perception-driven manipulation. A webcam or smartphone camera is used to detect the object position and orientation. Based on image processing and ARUCO marker detection, the robot calculates the required motion to grasp and place the object at the target location.

Working Procedure

1. Capture the workspace image using a camera.

2. Detect the object position using image processing.
3. Calculate object coordinates relative to the robot.
4. Move the robotic arm toward the detected object.
5. Grasp the object and move it to Point B.
6. Release the object at the target location.

Result

The robot successfully identified object locations and completed the pick-and-place operation using visual feedback.

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4. Task 3 – Object Handover Between Two Robot Arms

In this task, two robotic arms cooperate to transfer an object between their grippers. The first arm moves the object to a shared handover position while the second arm aligns its gripper, grasps the object securely, and completes the transfer without dropping it.

Working Procedure

1. First robotic arm grasps the object.
2. Move the object to the handover position.
3. Second robotic arm approaches the object.
4. Second gripper grasps the object securely.
5. First robotic arm releases the object.
6. Second robotic arm moves the object to the final position.

Result

The two robotic arms successfully coordinated the handover operation without dropping the object.

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5. Technologies Used

Component	Purpose

Braccio Robotic Arm	Object manipulation
Servo Motors	Joint movement control
Webcam / Smartphone Camera	Visual sensing
Python / Arduino	Robot programming and control
ARUCO Markers	Object localization

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6. Results and Conclusion

The project successfully demonstrated robotic object manipulation, camera-assisted perception, and multi-robot coordination. The implementation provided practical understanding of robotic kinematics, motion planning, grasping, and perception-assisted manipulation systems.

The project also improved knowledge of robotic programming, sensor integration, and robot coordination techniques used in modern automation systems.

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7. GitHub Repository Link

`https://github.com/Meetvirugama/robotics-project-ie410`

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8. Project Video Drive Link

`https://drive.google.com/file/d/1GrtE0H8TUWDI1zfG5Pl3ai5-VM1ELaU1/view?usp=sharing`